

File permissions in Linux

Project description

The overall purpose of the project is to help implement correct file permissions to help keep systems secure. Linux administrators can change these permissions for user, group, and others to read, write, and execute respectively. The task is to examine existing permissions on the file system and modify permissions to the correct authorization given.

Check file and directory details

```
researcher2@b2436c3823c5:~$ ls
projects
researcher2@b2436c3823c5:~$ cd projects
researcher2@b2436c3823c5:~/projects$ pwd
/home/researcher2/projects
researcher2@b2436c3823c5:~/projects$
```

Here is the command on how to go to the projects directory:

- ls – list all directories and files
- cd name – navigates to the chosen directory
- pwd – defines the current path of the working directory (hence, path working directory)

```
researcher2@b2436c3823c5:~/projects$ ls -la
total 32
drwxr-xr-x 3 researcher2 research_team 4096 Mar  7 05:05 .
drwxr-xr-x 3 researcher2 research_team 4096 Mar  7 06:53 ..
-rw-r--r-- 1 researcher2 research_team  46 Mar  7 05:05 .project_x.txt
drwx--x--- 2 researcher2 research_team 4096 Mar  7 05:05 drafts
-rw-rw-rw- 1 researcher2 research_team  46 Mar  7 05:05 project_k.txt
-rw-r----- 1 researcher2 research_team  46 Mar  7 05:05 project_m.txt
-rw-rw-r-- 1 researcher2 research_team  46 Mar  7 05:05 project_r.txt
-rw-rw-r-- 1 researcher2 research_team  46 Mar  7 05:05 project_t.txt
researcher2@b2436c3823c5:~/projects$
```

- ls -la – provides the list of permissions the file/s (both public and hidden) have in a current directory.

```
researcher2@b2436c3823c5:~/projects$ whoami
researcher2
researcher2@b2436c3823c5:~/projects$
```

As you can see, the name of the user is researcher2. While the group where researcher2 has access upon is research_team.

Describe the permissions string

```
researcher2@b2436c3823c5:~/projects$ ls -la
total 32
drwxr-xr-x 3 researcher2 research_team 4096 Mar  7 05:05 .
drwxr-xr-x 3 researcher2 research_team 4096 Mar  7 06:53 ..
-rw-r--r-- 1 researcher2 research_team  46 Mar  7 05:05 .project_x.txt
drwx-x--x 2 researcher2 research_team 4096 Mar  7 05:05 drafts
-rw-rw-rw- 1 researcher2 research_team  46 Mar  7 05:05 project_k.txt
-rw-r----- 1 researcher2 research_team  46 Mar  7 05:05 project_m.txt
-rw-rw-r-- 1 researcher2 research_team  46 Mar  7 05:05 project_r.txt
-rw-rw-r-- 1 researcher2 research_team  46 Mar  7 05:05 project_t.txt
researcher2@b2436c3823c5:~/projects$
```

The permissions string is comprised of 10 characters that dictate which owner has permissions to access/modify the file. Let's break it down:

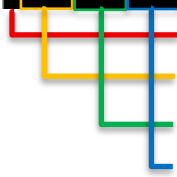
- d means directory (current path of where you are)
- r means read (can only view)
- w means write (can only write the contents)
- x means execute (can help save, delete, etc.)

Now in a file, there are three users: user, group, and others. Each user has three permissions: user has read, write, execute; groups has read, write, execute; and others has read, write, execute. Let's break it down as to why they're needed:

- u means users - the owners of the specific file and has the highest level of control
- g means group - set of users who share the specific file for collaboration and sharing
- o means others - everyone on the system that is not the owner or part of the group

Now that you know the differences and what these users and permissions do, let's decipher what the permissions string mean in a file called project_m.txt:

```
-rw-r--r-- 1 researcher2 research_team 46 Mar  7 05:05 project_m.txt
```

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- This is where d (stands for directory) is located
 - This section is for user (they have read and write permissions)
 - This section is for group (only has read permission)
 - This section is for others (has no permissions)

The overall conclusion here is that project_m.txt has read and write permissions from the user. While having only read permission from the group. Lastly it is private since it has no permission from others due to sensitive information.

Change file permissions

BEFORE

```
researcher2@b2436c3823c5:~/projects$ ls -la
total 32
drwxr-xr-x 3 researcher2 research_team 4096 Mar  7 05:05 .
drwxr-xr-x 3 researcher2 research_team 4096 Mar  7 06:53 ..
-rw--w---- 1 researcher2 research_team  46 Mar  7 05:05 .project_x.txt
drwx--x--- 2 researcher2 research_team 4096 Mar  7 05:05 drafts
-rw-rw-rw- 1 researcher2 research_team  46 Mar  7 05:05 project_k.txt
-rw-r----- 1 researcher2 research_team  46 Mar  7 05:05 project_m.txt
-rw-rw-r-- 1 researcher2 research_team  46 Mar  7 05:05 project_r.txt
-rw-rw-r-- 1 researcher2 research_team  46 Mar  7 05:05 project_t.txt
researcher2@b2436c3823c5:~/projects$
```

AFTER

```
researcher2@b2436c3823c5:~/projects$ chmod o-w project_k.txt
researcher2@b2436c3823c5:~/projects$ ls -la
total 32
drwxr-xr-x 3 researcher2 research_team 4096 Mar  7 05:05 .
drwxr-xr-x 3 researcher2 research_team 4096 Mar  7 06:53 ..
-rw--w---- 1 researcher2 research_team  46 Mar  7 05:05 .project_x.txt
drwx--x--- 2 researcher2 research_team 4096 Mar  7 05:05 drafts
-rw-rw-r-- 1 researcher2 research_team  46 Mar  7 05:05 project_k.txt
-rw-r----- 1 researcher2 research_team  46 Mar  7 05:05 project_m.txt
-rw-rw-r-- 1 researcher2 research_team  46 Mar  7 05:05 project_r.txt
-rw-rw-r-- 1 researcher2 research_team  46 Mar  7 05:05 project_t.txt
researcher2@b2436c3823c5:~/projects$
```

As you can see, the command `chmod o-w project_k.txt`:

- Chmod – change mode for the permission
- o-w – set the others to delete write permission (hence other minus write)
- project_k.txt – the name of the file to have its permissions modified.

Therefore, the command specifies that the `project_m.txt` file has been modified to exclude others to have permission to write the file.

Change file permissions on a hidden file

```
researcher2@b2436c3823c5:~/projects$ chmod o-w project_k.txt
researcher2@b2436c3823c5:~/projects$ ls -la
total 32
drwxr-xr-x 3 researcher2 research_team 4096 Mar  7 05:05 .
drwxr-xr-x 3 researcher2 research_team 4096 Mar  7 06:53 ..
-rw--w---- 1 researcher2 research_team  46 Mar  7 05:05 .project_x.txt
drwx--x--- 2 researcher2 research_team 4096 Mar  7 05:05 drafts
-rw-rw-r-- 1 researcher2 research_team  46 Mar  7 05:05 project_k.txt
-rw-r----- 1 researcher2 research_team  46 Mar  7 05:05 project_m.txt
-rw-rw-r-- 1 researcher2 research_team  46 Mar  7 05:05 project_r.txt
-rw-rw-r-- 1 researcher2 research_team  46 Mar  7 05:05 project_t.txt
researcher2@b2436c3823c5:~/projects$
```

As stated from this screenshot, the hidden file (starts with .) referred to as “.project_x.txt” is accessible by users and groups. The goal is to include read access to user and group and exclude write permissions. Here is the solution to the problem:

```
researcher2@b2436c3823c5:~/projects$ chmod ug+r .project_x.txt
researcher2@b2436c3823c5:~/projects$ chmod ug-w .project_x.txt
researcher2@b2436c3823c5:~/projects$ ls -la
total 32
drwxr-xr-x 3 researcher2 research_team 4096 Mar  7 05:05 .
drwxr-xr-x 3 researcher2 research_team 4096 Mar  7 06:53 ..
-r--r----- 1 researcher2 research_team  46 Mar  7 05:05 .project_x.txt
drwx--x--- 2 researcher2 research_team 4096 Mar  7 05:05 drafts
-rw-rw-r-- 1 researcher2 research_team  46 Mar  7 05:05 project_k.txt
-rw----- 1 researcher2 research_team  46 Mar  7 05:05 project_m.txt
-rw-rw-r-- 1 researcher2 research_team  46 Mar  7 05:05 project_r.txt
-rw-rw-r-- 1 researcher2 research_team  46 Mar  7 05:05 project_t.txt
researcher2@b2436c3823c5:~/projects$
```

Where I utilized the command “chmod ug+r .project_x.txt” to let user and group have read permissions. I also utilized “chmod ug-w .project_x.txt” to delete write permission/s from user and group. The results from the screenshot showed .project_x.txt has only read permissions from user and group.

Change directory permissions

```
researcher2@b2436c3823c5:~/projects$ chmod g-x drafts
researcher2@b2436c3823c5:~/projects$ ls
researcher2@b2436c3823c5:~/projects$ ls -la
total 32
drwxr-xr-x 3 researcher2 research_team 4096 Mar  7 05:05 .
drwxr-xr-x 3 researcher2 research_team 4096 Mar  7 06:53 ..
-r--r----- 1 researcher2 research_team  46 Mar  7 05:05 .project_x.txt
drwx----- 2 researcher2 research_team 4096 Mar  7 05:05 drafts
-rw-rw-r-- 1 researcher2 research_team  46 Mar  7 05:05 project_k.txt
-rw----- 1 researcher2 research_team  46 Mar  7 05:05 project_m.txt
-rw-rw-r-- 1 researcher2 research_team  46 Mar  7 05:05 project_r.txt
-rw-rw-r-- 1 researcher2 research_team  46 Mar  7 05:05 project_t.txt
researcher2@b2436c3823c5:~/projects$
```

The goal of this task is to change directory permissions so that only user (researcher2) has the rwx (read, write, execute) permissions within the “drafts” directory. The solution is the same as the previous ones. Let’s break it down to understand the answer:

- `chmod g-x drafts` – change access of the group’s execute permission for the directory “drafts”.

The results showed that only user has rwx permissions within “drafts”. Therefore, leaving groups and others unable to access/modify the contents inside the “drafts” directory.

Summary

The overall project provided a step-by-step tutorial on how to change and modify permissions for users in a file/s or directory. With permissions that can alter the contents of the file or directory, this is related to the overall security principles on letting employees have access to only tools and files depending on the responsibility of their roles.